(cc4) Decision Tree and Random Forest

Q1) which measure is 0 for a pure node? C) Both

Q2) Entropy of a node with 4 pass and 4 fail? C) 1

Q3) Gini of a node with 6 pass and 2 fail? Gini: $1-((\frac{6}{8})^2+(\frac{2}{8})^2) = .375$

B) .375

Q4) The best split is the one with: B) Highest gain

Q5) H(parent) = .954, entropy = .406 Gain? Gain = .954 - .406 = .548

B) .548

Q6) Train accuracy 1.00 but test accuracy .80 suggests:

B) overfitting

Q7) unique rows in a bootstrap sample (large n):

C) about 63.2%

Q8) out-of-bag rows are used to: B) Estimate error for free

Q9) Votes of 7 trees: 4 vote yes, 3 vote no. Result?

$P_{yes} = 4/7 = .57$

B) yes, .57

Q10) A random forest mainly reduces: B) variance